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DEVELOPMENT OF AN INTEGRATED HARDWARE PLATFORM FOR AUTONOMOUS ROBOT NAVIGATION

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autore della presente tesi di Laurea dal titolo
DEVELOPMENT OF AN INTEGRATED HARDWARE PLATFORM FOR AUTONOMOUS
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Parola chiave Robotics, Hardware Design and Integration, Autonomous Robot, Ros

Abstract Hardware design and development of a tank-robot for autonomous navigation:
power supply system, control and computing devices, on-board sensors and
software integration.

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la consultazione della presente tesi, fatto divieto a chiunque di riprodurre in tutto o in parte quanto in
essa contenuto.

Bari,

Firma

Germano Pansini



To the ones,
who believed in me.



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1 Introduction

1.1 Abstract

How are robotics technologies shaping today's world? What are mobile robots and which tasks they must comply to be called autonomous? How to build a simple custom mobile robot? What an integrated hardware system consists of and what are the most common devices and tools that help us developing custom robots that can build a 3D map of the surrounding environment?

We're going to discuss these topics within this bachelor thesis, which presents a methodology for designing and developing mobile robotic systems. The design procedure taken in this development focuses on integrated hardware platform for robot navigation, but the knowledge achieved from it opens doors for the whole expanding field of robotics and mechatronics.

1.2 Internship

This thesis investigates on how we have designed a Hardware platform during my internship's research at the **SST Laboratory** in Polytechnic of Bari, in collaboration with a company based in Milan, Italy, called "E-viewing Technology".



Figure 1 SST Laboratory Logo



Figure 2 E-Viewing Technology S.r.l Logo

We started from scratch with an empty crawler of an existing tank robot that goes by the name of "Vantage Patrol Robot", created by an American company based in San Diego, CA, by the name of "Transcend Tactical", which I had to "give life" to.



Figure 3



Figure 4

Figure 3 and 4 representing SWAT using Transcend Vantage Tactical Robot

1.3 Goal of the thesis

The main project goal refers to hardware skills achieved during the Design and Development of my first hardware platform for mobile robots, aimed for Navigation tests.

This thesis contains a summary collection of knowledge and experiences on the processes that took place in order to build our robot, with the purpose of targeting which are the first steps I've followed in this kind of project. Of course, it is not our intention to just collect the experiences as set of instructions, but rather interpreting them as a starting point for future research and develop. As a matter of fact, the nature of this thesis wants to be more practical than theoretical, precisely because all the acquired knowledge about robotics are the results of studies and tests carried out by our team in the SST Laboratory.

At the moment, our robot navigation is controlled by human from a remote client station while it's autonomously creating a 3D point-cloud map of the surrounding environment due to the open-source code by the name SSL-SLAM, running on ROS. SSL-SLAM stands for Solid-State LiDAR Simultaneous Localization and Mapping and represent a common framework in mobile robotics that allows robots to localize themselves while mapping the environment using a particular type of sensors, LiDAR sensors. The Light Detection and Ranging (LiDAR) is a kind of Time of Flight (ToF) sensor which measures the object distance by emitting laser to the object and capturing the travelling time.

This algorithm runs on ROS (Robot Operative System), a set of software libraries and tools that help build robot applications, we are going to discuss in a dedicated chapter.

Anyway, it will be able to autonomously move in an unknown environment without any human interaction or significant change in the hardware platform we developed, but only in the Software integration, on which we are not focusing deeply this thesis. This is the main reason why thesis title still stands, even if autonomy isn't achieved yet.

1.4 Context of the Work

The hardware side consists in Designing and Developing a custom embedded system for mobile robot to perform specific tasks and function, as defined, in the field of navigation and this is the main process we are going to discuss in this thesis.

The design philosophy we will use for explaining the hardware integration of this robot is called "black box design", a common approach in engineering.

A black box refers to a system whose behaviour has to be observed entirely by inputs and outputs: we can see every single component, or group of them, as boxes and each of these boxes have predefined inputs/outputs with their own function, but it is not defined what actually is inside them. It is clear that this type of design leads us to better understand how different components are connected and what function they serve, allowing us to have a general view of the project without worrying about every single device, as the metaphor

components sharing data don't have to know anything about each other than the fact they have common interfaces.

(Heusser, s.d.)

In this way our robot project can be broken into three different layers: the **Navigation** Layer, **Intelligence** Layer, and the environmental **Inputs/Outputs**, we are going to explain further.

1.5 Organization

The rest of this thesis is organised as follows:

- Section **2**: provides a general Background and related Works on Mobile Robotics state of art.
- Section **3**: describes the Design and Development processes.
- Section **4**: is the core of the thesis and explain the integrated hardware Platform developed in the SST Laboratory.
- Section **5**: is dedicated to Tests and Experimental Results, mainly on **SSL-SLAM** using **ROS**.
- Section **6**: Conclusions and future Works.

2 Robotics in our Society

2.1 Definitions of “Robot” and “Robotics”

“Robotics is a subject
with shifting boundaries.”
(Todd, 1986)

Robotics is an interdisciplinary branch of computer science and engineering, whose objective is to design, construct and use machines that can perform high-level tasks done traditionally by human beings, in a language resembling those used by humans to communicate among themselves.

This subject integrates different fields such as electrical engineering, mechatronics, electronics, computer engineering, mechanical engineering, bioengineering, mathematics, and many aspects of robotics are involving today more and more artificial intelligence.

Anyway, we can say that “... the boundaries of Robotics are not only vague but shifting. Robots are evolving quickly, and our ideas with them...” (Todd, 1986). Just looking at Robotics etymology we can try to grasp its nature’s complexity, that involve scientific but art’s field as well.

As matter of fact the word “Robotics” was used for the first time in print by Isaac Asimov, an American writer and biochemistry professor

at Boston University, in his science fiction short story “Liar!”, published in May 1941 in Astounding Science Fiction.

Ironically Asimov was unaware he was coining the term, he just looked at the word *Electronics*, referring to the science and technology of electrical devices, and assumed that *Robotics* was already referring to the technology science of robots. In addition, *Robotics* was derived from the term *Robot*, which was introduced to the public in 1920 by another science fiction writer, Karel Čapek, in his play “*R.U.R (Rossum’s Universal Robots)*”, where robots were artificial people created from synthetic organic matter, very similar to the modern ideas of androids. The term *robot* comes from the Slavic word *robota*, which means work. Cfr. (Wikipedia, s.d.)

“... It means that each factory will be making Robots of a different colour, a different nationality, a different tongue; they’ll all be different – as different from one another as fingerprints...” (Čapek, 1920)



Figure 5 Liar from Isaac Asimov

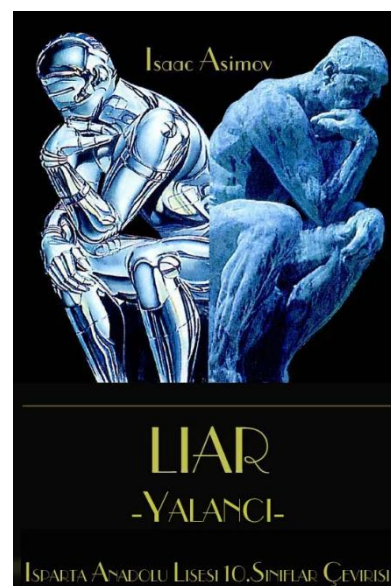


Figure 4 R.U.R from Karel Čapek

2.2 Differences between Robots and Machines

The fact that we are nowadays calling robots a large variety of machines, different in size, design and capabilities, makes it not so easy to come up with a clear and unique definition of what a robot is and what are the main differences between robots and just machines. Even roboticists have different conceptions about what type of machines can be defined robots and what type can't.

We have to keep in mind that both robot and machines can operate up independently to a certain extent and also be automated in order to run with minimal human control.

We can see **machines** as mechanical and/or electrical devices that can repeatedly complete a pre-determined motion and task without the capability of *thinking* and acting on their own to respond to changes in external environment, like fork lifters.

A **robot** can do that too but has also an extra class of functionality to an equivalent machine, because of its functions approaching intelligence given by, at least, some computer components: it can be programmed in different ways to perform many different tasks, with little human control or without, having some sort of awareness to certain degree depending upon its type.

We can say that all robots are machines but not all machines are robots.

There are plenty of differences between a machine and a robot, but in the following we are going to underline the main characteristics that let us define a machine a robot:

Autonomy

It is seen as the robot capability of performing its own functions with little or no direct human control. The machine relies, more than robot, on human's interaction and supervision to work. Humans constantly have to monitor and control processes with machine, while robot needs only a set of instructions detailing how the job should be done. For this reason, we can see a normal car as a typical machine, and an autonomous one as a robot.

Sensory Input

Robots take advantage of sensory data to manage their performance while in machine this is almost totally absent, except for a small class. Machines usually rely on an external trigger to work, as we said before. Electronic communication devices, such as cameras and microphones, help transmit sensory data to computers simulating human nervous system. This is fundamental to allow robots interaction with the environment.

Software

This plays an important role in differentiating between machines and robots. If they do, machines have simple software, whereas robots need advanced one. Without software, a robot is a piece of hardware



just like any other machine. This is the actual “brain” of a robot, and this is where programming enters the picture.

Design and Development

There's a degree of rigidity in the machine function, instead designing and developing robots it's wide and more complex. That's because robots are mostly designed to operate autonomously for certain periods. They not only require more in terms of research and development for their functionality, but they also need to be tested extensively to ensure they match the performance parameters intended for them.

Performing changing Tasks

Robots can be programmed to fulfil different tasks and are more prone to configuration changes. So, robots are in a way more versatile.

Artificial Intelligence

Approach based on artificial intelligence have become more and more relevant into the field of robotics. With this, a robot can be capable of learning and adapting to different conditions in a more flexible way than before. Artificial intelligence and robotics interact through software.

(White, s.d.)

2.3 How do Robots benefit Society?

Due to their abilities, mobile robots find uses in multiple areas, especially where they can alleviate strenuous tasks and complete missions that are too dangerous for a human to undertake.

(E.g.: first Responders, Planet Exploration, outdoor Security and Inspections, Automotive and so on)

Let's have a look to some of the advantages robots bring in the society:

Safety

The first industrial robot, called, was introduced to the U.S. public in 1962 by George Charles Devol and Joseph Frederick Engelberger, who would come to be known as “the father of robotics”.



Figure 6 Charles Devol and Joseph Frederick Engelberger

“This robot undertook the job of transporting die castings from an assembly line and welding these parts on auto bodies, a dangerous task for workers, who might be poisoned by toxic fumes or lose a limb if they were not careful.” (Wikipedia, Unimate, s.d.)

Since that time the first benefit to society obtained with automation and robotics, safety in the workplace, hadn't change at all. In fact, robots can handle lifting heavy loads, toxic substances and can perform repetitive tasks and, more, delegating them this type of dangerous task, remove some of the risks taken before by employees. Our planet is getting increasingly vulnerable due to both natural and man-made disasters and these types of events are unavoidable in most of the case, so it's of vital importance to identify how we can prevent disasters or help victims and save lives. In extreme environments, such as for inspection and decommissioning of nuclear sites, the development of robotic technology has received considerable attention in recent years.

(ESA , s.d.)

In some other cases they let us do things that aren't only dangerous but rather impossible for our technology level, as travelling to Mars. In fact, this would require scientist and engineers to overcome a range of technological and safety obstacles, as for instance, radiation taken from the sun.

Thanks to robot development, from the evening of August 5, 2012, the rover Curiosity continues to explore today Mars land, having taken more than 700,000 raw images and travelled over 14 miles on its surface.



Figure 7 Curiosity on Mars Land



Production Speed

It seems obvious that, unlike human beings, robots don't need breaks or get distracted and can work all the time until low battery's advice. In the field of high-pressure works, they can easily speed up the production making company's growth possible and easier.

Quality and Consistency

Automation is typically far more reliable than human labour in some sectors. Since they are programmed for repetitive and precise motion, they're less likely to make mistakes and in most of the case robots are simultaneously employees and quality control system.

Job Creation

We have to understand that robots don't take jobs away but instead they change them: in fact, robots still need people for monitoring and supervision and the more robots we need, the more people we'll need to build those robots.

This represents an opportunity to growth as society to a level never seen up to now.

2.4 What is Autonomous Navigation?

Navigation is the ability to determine your own location within an environment and to be able to figure out a path that will take you from your current location to some final destination.

Navigating in the wilderness might require for example GPS to determine where you are and a map to plan the best path to get around mountains, lakes to reach your camp site or whatever your goal is. Autonomous navigation is doing exactly this but without a human in the loop.

Broadly speaking it's how we get a vehicle to determine its location using a set of sensors and then to move on its own through an environment to reach a desired goal. We use the term vehicle referring on any kind of machine: it could be a car travelling down a road or a UAV making its way back to the airport or a spacecraft journeying across the solar system or a submersible exploring the depths of the ocean or some other mobile robot.

In this way we've given the vehicle autonomy: the ability to make decisions and act on its own.

But there are different levels of autonomy, and they range from a vehicle that is simply operated by a human from a remote location, but it has some simple algorithms onboard that will take over and autonomously keep it from running off a cliff or something, all the way up to fully autonomous vehicle with no human interaction at all.

3 Design and Development Process

Mobile robotics is nowadays one of the fastest expanding sectors of scientific research with the main goal of extending the exploration capabilities of humans, by providing access to dangerous structure, extreme environments or unreachable areas.

In this section we are going to give a basic understanding of mobile robotic, focusing deeply into the design process to explain what the common steps are, if we want a custom mobile robot for robot exploration. The design and construction of a mobile robot incorporates mechanical and electrical engineering along with some computer programming.

The design procedure can be defined as an iterative decision-making process to conceive and implement optimum systems to fulfil specific task.

The goal of the project is providing a versatile platform for our future research in the field of mobile robotics.

The development process of our custom robot is based on the following steps:

- **3.1 Defining Project Requirements and Objectives**
 - **3.2 Project Planning**
 - **3.3 Mobile Robotics Layers**
 - **3.4 Component Selection**
-

- 3.5 Soldering and Assembly
- 3.6 Design and Development of Subsystems
- 3.7 Integration of the Hardware Platform
- 3.8 Experimental Results and Testing

3.1 Defining Project Requirements and Objectives

Defining project requirements and objectives is the first essential step to take, even before beginning a project. If objectives aren't clear at the beginning, the final product can turn up being different from the one which was required at the initial level. We have to clarify project planning, schedule phases and outline development strategies at early stage. Doing this, not only save us an important amount of time and money but allows to keep our work constantly in compliance with our goal.

The Requirements and Objectives of our project are stated as below:

- Navigation of robot shall be controlled from a remote station using a USB Joystick firstly, and then will be completely autonomous.
(This last task is not achieved yet, but it relies more on software development which is not the main core of this thesis).

- Location and destination points are specified by user.
- The mobile robot shall be designed for rough and loose terrain but for smooth surfaces too.
- Robot has to climb obstacles and stairs.
- Standard speed of robot shall be specified by user.
- Processors, sensors and actuators shall be on-board.
- Power system for motors and electronic devices required on-board.
- Robot must be sufficiently powered to carry 10Kg payload.
- Using SSL-SLAM open-source code, robot shall be able to build and update 3D Map of the surrounding environment, while understanding its location.

3.2 Project Planning

When we've started working on this project, the capabilities of carrying heavy payload, climbing obstacles and stairs and being able to move through any kind of surfaces were enabled as the robot chassis and its skid locomotion system was already selected and integrated.

The product specifications consider the other defined requirements and objectives. Microcontroller, motor driver and an on-board computer equipped with LiDAR were intended to be used for performing SSL-SLAM (a dedicated discussion in the 5th Chapter). After the component's selection, we had to solder, assembly and integrate on-board, connecting all the required hardware devices to

allow robot first movements. As Power sources for Motors and on-board Computing devices we decided to use rechargeable 22.2V DC LiPo Batteries of 5Ah and 10Ah, based on the calculated system power requirements.

3.3 Mobile Robotics Layers

A mobile robot is a machine controlled by software that uses sensors and other technology to identify its surroundings and move around its environment, whose design must be based on 4 main interconnected layers:

3.3.1 Locomotion

3.3.2 Perception

3.3.3 Cognition

3.3.4 Navigation

We are going to briefly explain these layers, without going into details. To analyse the structure of our robot we're going to develop the first layer, the locomotion subsystem.

3.3.1 Locomotion

The locomotion subsystem consists of methods that robots use to transport themselves from place to place and is the first issue that affects mobile robot structure in the design process.

In fact, there are a large variety of possible ways to move: we have **stationary** robots, when the robot's base is fixed and they consist of an open kinematic chain such as arm manipulators and industrial robots, capable of moving objects applying force to them, **walking** robots, that simulate human or animal gait using a small number of articulated legs, **rolling** robots, when their structure include wheels or a number of continuous tracks simulating cars or tanks, but also more complex ones like **hopping**, **slithering**, **swimming** or **hybrid** robots, we are not going to focus on. Generally legged locomotion requires higher degrees of freedom and greater mechanical complexity than wheeled locomotion.

The selection of the locomotion system depends strictly on the type of environment our mobile robot will be working in.

Wheels, in addition to being simple, are extremely well suited to flat and hard ground but as the surface becomes soft, wheeled locomotion accumulates inefficiencies due to rolling friction, whereas legged locomotion suffers much less, because it has only few contact points with the ground.

We can see at locomotion as the complement of manipulation: while in manipulation the robot arm is fixed and applies force to the objects around it, instead in locomotion the environment is fixed, and the robot can move by applying force to it.

(Roland Siegwart, Illah R Nourbakhsh, 2004)

Anyway, we will not delve into the physical basis of locomotion because this thesis focuses more precisely on the development of an integrated hardware platform for **tracked** robot navigation.

3.3.1.1 Rolling robots (Tracked slip/Skid Locomotion)



Figure 8 Our robot Skid Locomotion subsystem

As we said before, among rolling robots are being included both wheeled and tracked robot. They have been by far the most popular locomotion mechanism in mobile robotics, and in man-made vehicles in general, because they can achieve good efficiencies with a relatively simple mechanical implementation.

For example, with rolling robots, we don't have to worry about balance because all the tracks, or wheels, are always designed to be always in ground contact, instead we must focus our research on the problems of wheel geometry, stability, manoeuvrability, traction, and control.

Our robot was already equipped with Tracked slip/Skid locomotion. Using treads for the locomotion system of our robot significantly improve the manoeuvrability in loose terrain, compared to the traditional wheeled robots, however, due to this large ground contact patch, changing the orientation of the robot usually requires a skidding turn, wherein a large portion of the track must slide against the terrain. The exact centre of rotation of the robot is hard to predict

because of the large amount of skidding during a turn. This type of system works perfectly over rough and loose terrain.

3.3.1.2 Key Issues for Locomotion

In the design of Locomotion system, we have to pay particular attention to many aspects. Examples in the following bullet points:

- Stability
 - Centre of gravity
 - Contact points
 - Inclination

- Type of environment
 - Presence of undesired objects
 - Type of terrain

- Friction

- Wheel geometry or slipping track design

3.3.2 Perception

Perception consists in acquiring knowledge about the environment our robot is operating in and this represents one of the most important tasks that a mobile robot system requires.

How can a robot do this? This is typically done taking measurements using various sensors and then extracting meaningful information from them. The variety of sensors used in mobile robots are developing so fast that is quite impossible to focus on each type. Some sensors measure simple value like the internal temperature of the robot electronics or RPM of the motors, other, more sophisticated, can be used to acquire information about robot's environment or position. In our project the only sensor we've used by now is LiDAR.

I'll be explaining more about what LiDAR is and how it works in subsection 4.7.



Figure 9 General Perception schematic

3.3.2.1 Sensor Classification

We give a simple classification of sensors type below:

- **Proprioceptive**

These sensors measure values internal to the robot system.
(Motor speed, Wheel Load, Battery Voltage, etc)

- **Exteroceptive**

These sensors acquire information about robot's environment.
(Distance Measurement, Light Intensity, Sound Amplitude, etc)

- **Passive**

They measure ambient environmental energy entering the sensor.
(Temperature probes, Microphones, CCD/CMOS Cameras)

- **Active**

They emit energy towards the surrounding environment and then measure the environmental reaction, they can manage more interactions, and often achieve superior performance.
(Wheel quadrature encoders, Ultrasonic Sensors, and Laser Rangefinders such as our LiDAR)

3.3.2.2 Sensor Performance

When using sensors, we have to keep in mind at least some of the main parameters given below:

- **Sensitivity**

This is a measure of the degree to which an incremental change in the target input signal changes the output signal.

It is the ratio of the output variation to the input variation.

- **Error**

This is defined as the difference between the sensor's output measurements and the true value being measured.

Given a true value n and a measured one m this is

$$Error = m - n$$

- **Accuracy**

This is the degree of conformity between sensor measurement and the true value, often expressed in terms of percentage of the true value.

$$Accuracy = 1 - \frac{|Error|}{n}$$

- **Systematic Errors**

They are caused by factors/processes that can be modelled in theory.

Possible examples of cause of systematic errors can be poor calibration of a laser rangefinder, unmodeled slope of a hallway floor, bending of stereo camera head due to collision.

- **Radom errors**

These types of errors cannot be predicted using a model and can only be described in probabilistic terms. Examples: hue instability in a colour camera, spurious range finding errors...

- **Precision**

Often confused with accuracy, this relates to reproducibility of the sensor results. A sensor taking multiple readings of the

same environmental state has high precision when it produces the same output. This property is distinct from the accuracy of the sensor output.

$$Precision = \frac{Range}{\sigma}$$

Next picture list common sensors used in mobile robotics applications, classified according to their use, system and fundamental characteristic seen before.

General classification (typical use)	Sensor Sensor System	PC or EC	A or P
Tactile sensors (detection of physical contact or closeness; security switches)	Contact switches, bumpers	EC	P
	Optical barriers	EC	A
	Noncontact proximity sensors	EC	A
Wheel/motor sensors (wheel/motor speed and position)	Brush encoders	PC	P
	Potentiometers	PC	P
	Synchros, resolvers	PC	A
	Optical encoders	PC	A
	Magnetic encoders	PC	A
	Inductive encoders	PC	A
	Capacitive encoders	PC	A
Heading sensors (orientation of the robot in relation to a fixed reference frame)	Compass	EC	P
	Gyroscopes	PC	P
	Inclinometers	EC	A/P
Ground-based beacons (localization in a fixed reference frame)	GPS	EC	A
	Active optical or RF beacons	EC	A
	Active ultrasonic beacons	EC	A
	Reflective beacons	EC	A
Active ranging (reflectivity, time-of-flight, and geometric triangulation)	Reflectivity sensors	EC	A
	Ultrasonic sensor	EC	A
	Laser rangefinder	EC	A
	Optical triangulation (1D)	EC	A
	Structured light (2D)	EC	A
Motion/speed sensors (speed relative to fixed or moving objects)	Doppler radar	EC	A
	Doppler sound	EC	A
Vision-based sensors (visual ranging, whole-image analysis, segmentation, object recognition)	CCD/CMOS camera(s)	EC	P
	Visual ranging packages		
	Object tracking packages		

A, active; P, passive; P/A, passive/active; PC, proprioceptive; EC, exteroceptive.

Figure 11 Classification of sensors used in mobile robotics applications

3.3.4 Cognition

One of the most challenging capabilities required in a mobile robot is Navigation. Succeeding in navigation requires effectiveness of the following four building blocks:

- **Perception**

Robot must interpret its sensors, in order to be able to extract meaningful data.

- **Localization**

The ability to determine its position in the surrounding environment.

- **Cognition**

The robot must be able to decide how to act to achieve its goals.

- **Motion Control**

The robot must properly control its motors to achieve the desired trajectory.

Of these four, localization has received great research attention in the past decade and is our main subject in this section.

We present the general block diagram of the localization in the following figure:

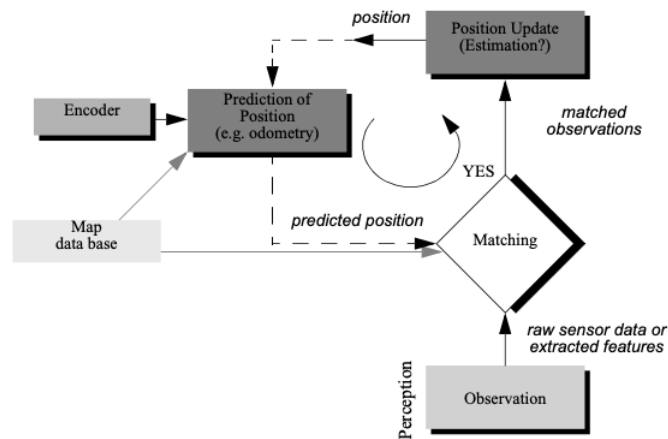


Figure 10 General Localization system block diagram

In this layer we are turning our attention to the robot cognitive level, which represents the purposeful decision-making and execution that a system utilizes to achieve its highest-order goal.

In our case of a mobile robot, the aspect of cognition directly linked to a robust mobility is navigation competence.

“Cognitive robots achieve their goals by perceiving their environments, paying attention to the events that matter, planning what to do, anticipating the outcome of their actions and the actions of other agents, and learning from the resultant interaction.”

(IEEE Robotics & Automation Society, s.d.)

During the cognition step, a robot has to select a strategy for achieving its goals. If its goal is to reach a particular location, the robot may need to acquire or build an environmental model, a map, that aids it in planning a path to the goal. Localization, in fact, doesn't mean only determining an absolute position of the robot in space but it means building a map and then identifying the robot position relative to that. Robot sensors and actuators play a fundamental role.

We have to pay attention also to the following sensor parameters:

- Sensor Noise
- Sensor Aliasing
- Actuator Noise

If we represent the position of a robot with a 3 components vector $p = (x, y, \theta)$, it is important to establish error model for position estimation to understand how errors propagate over time.

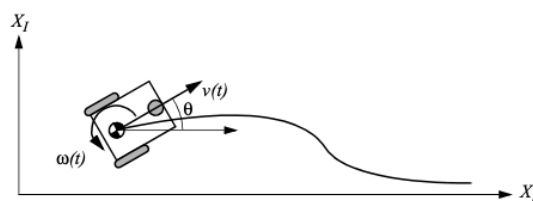


Figure 11 Movement of a differential-drive Robot

3.3.5 Navigation

Given partial, or total, knowledge about the environment, the goal position or a series of goal positions, navigation consists in the ability of the robot to act in order to reach its goal positions as reliably as possible. (Siegwart & Nourbakhsh, 2004)

Many approaches have been proposed for solving the navigation problem in the mobile robotics research community.

The main difference between various navigation architectures is the manner in which they decompose the problem into smaller subunits. Additional competences required for mobile robot navigation are:

- **Path Planning**

Involves identifying a trajectory that allows robot to reach goal location. This is a strategic problem-solving competence. Robot must decide what to do over long term to achieve its goals. The environment representation can range from a continuous geometric description to a decomposition-based geometric map or even a topological map. The first step is to transform the continuous environmental model into a discrete map.

Common road map and cell decomposition path-planning techniques are listed below:

- Road map path-planning
- Visibility graph
- Voronoi Diagram

- Cell decomposition path-planning
- Exact cell decomposition
- Approximate cell decomposition
- Potential field path planning

- **Obstacle Avoidance**

Given real-time sensors measurements, this means modulating the trajectory of the robot to avoid collisions.

Some of the techniques are:

- The Bug Algorithm.
- The Bubble Band
- The basic Curvature Velocity approach
- The lane Curvature Method
- Dynamic Window Approaches
- The Schlegel Approach
- The ASL

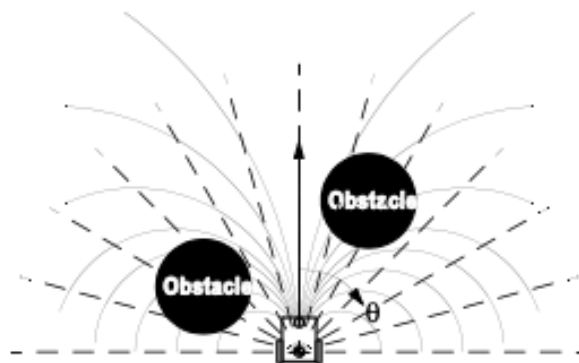


Figure 12 Robot scheme for Obstacle Avoidance

3.4 Components Selection

Availability increase of commercial components in vast varieties and categories has facilitated system development processes.

Our component choice is based on the project plan:

- Two rechargeable LiPo Battery of 22.2V DC and 10Ah/5Ah
- LiPo battery Tester and Charger
- Arduino UNO R2
- Fanless Computer
- LiDAR L515 from *IntelRealSense*
- USB Joystick from *Logitech*
- Sabertooth 2x60 Dual Motor Driver
- Plug connectors (XT-90, XT-90s, XT-60, Power Jack)
- ON-OFF Toggle Switch
- Different AWG Wire and Heat-Shrink Tubes
- Stripping Plier
- DC-DC buck Converter



Figure 13 Some components of the list (LiPo batteries, balance/power Cables and Charger)

3.5 Soldering and Assembly

Before assembling components, we need to know how all devices are going to be integrated and connected to each other. Next step consists in soldering different type of connectors, enabling wiring compatibility among the large variety of input/output ports of the electronics on-board. Examples given below:

- LiPo batteries female plug and LiPo Charger didn't share same connector type, so we had to unsolder and solder again, enabling the charging process.
- Build wires with different AWG size and then solder required plug connector for all electronics onboard (Arduino, Motor Driver, Power system for the on-board Computer).
- Create an ON-OFF toggle switch system

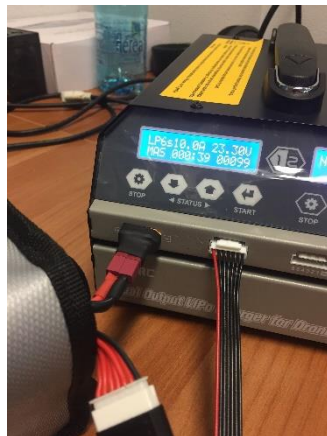


Figure 15 LiPo battery charging Process



Figure 14 Custom Cables

All soldering process was made by me at the SST Laboratory.

3.6 Design and Development of Subsystems

In the development and integration of a hardware platform for a mobile robot, it's typical to break down the project into three different layers:

1) **Navigation** layer:

In our case it consisted in the implementation of Chassis, Motors, Motor Controllers, Drivers, Sensors and the Power Distribution System.

All these have allowed the robot to perform its movement.

2) **Intelligence** layer:

When implementing a computer onboard, programmed to receive and transmit data to perform specific tasks, we give robot a "brain", which can control the Navigation layer working with the different onboard Hardware devices, thanks to its Software library and tools such as ROS.

It allows the robot to analyse and compute data received from sensors and, for our project, to build a 3D map of the unknown environment. The Intelligence layer also performs the master communications with the outside world.

3) **Environmental Inputs & Outputs** layer:

This consists of all the devices that can sense the local environment and collect data from it. These are high-level sensors which are able to pre-process data, making it easier for the intelligence layer to

work with them. In our case we worked with LiDAR, Cameras and with the Joystick Controller.

A black-box representation of the different subsystems is given below.

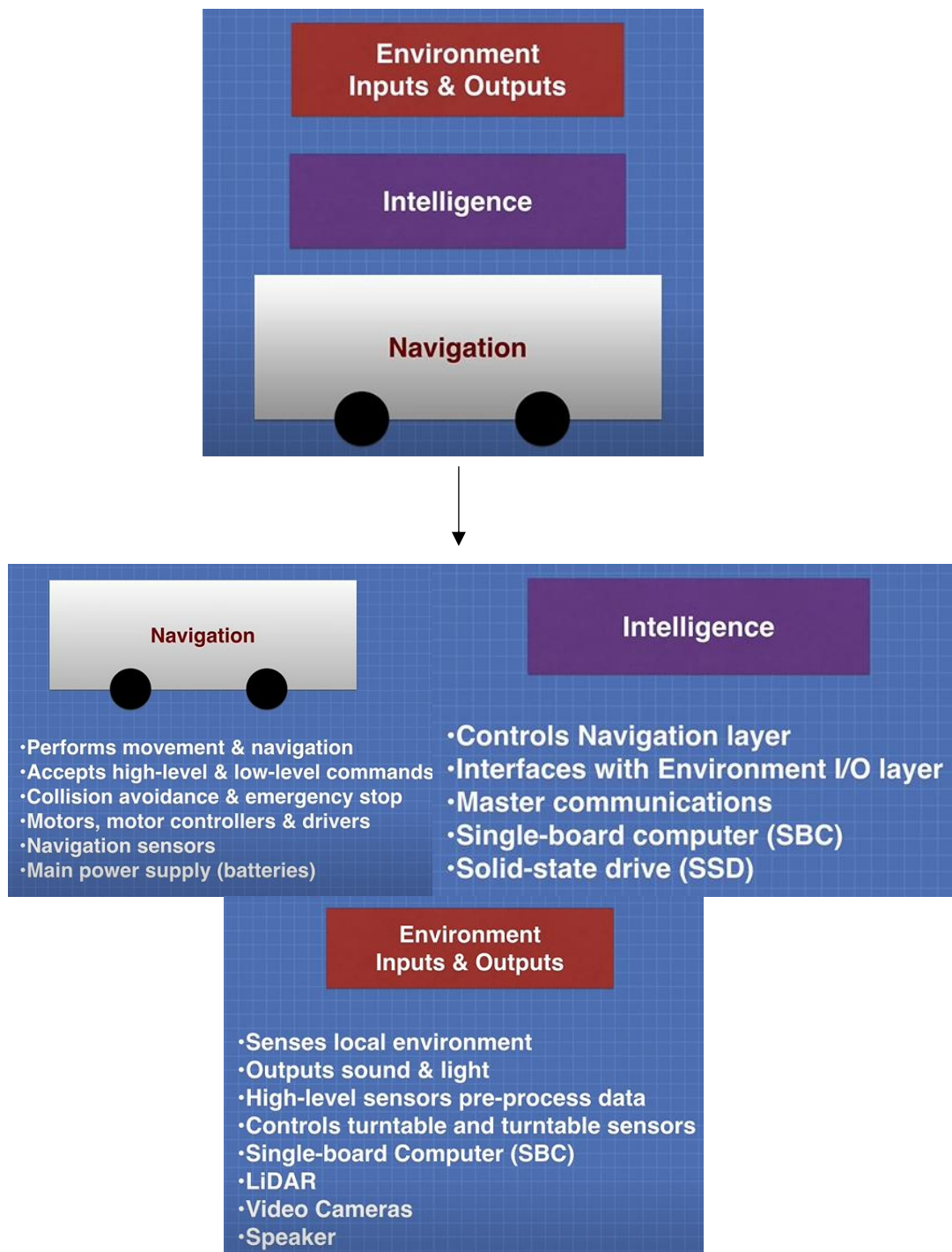


Figure 16 Subsystem Design (Navigation, Intelligence and Environmental Inputs/Outputs layers)

3.7 Integration of the Hardware Platform

3.5 and 3.6 are better developed in the following Chapter 4, as it has actually been my practical experience at the SST-Laboratory during my internship and we want to dedicate an entire chapter to these processes.

After understanding how the different subsystem composes the entire robot, what type of electronic devices and sensors have to be connected together and how they communicate with each other, the next step is the integration of all of them in a single hardware platform inside the robot structure.

3.8 Experimental Results and Testing

To solve problems of integration, our subsystems were made compatible with each other.

The interfaces of each subsystem were tested and then validated before and after integration processes. We tested step by step the integrated system assembly, each time a subsequent subsystem was linked with the previous one. After assembly and integration, testing at system level was done to verify that system is fulfilling its requirements.

(Tareen, 2016)



Experimental Results aim to test:

- **On-board Powering Systems**
- **SSH**
- **Remote-Control Navigation**
- **SSL-SLAM** (More about in subsection 4.8)

Chapter 5 is dedicated to discussing about experimental Results and Testing.

4 Development of an Integrated Hardware Platform

4.1 Introduction

This is the core section of my thesis and consists in Design and Development of an Integrated Hardware Platform for an empty crawler of an existing tank robot, that goes by the name of *Vantage Patrol Robot*, developed by an American company based in San Diego, CA, called *Transcend Tactical*.

An Italian company based in Milan, IT, by the name of *E-Viewing Technology*, gave it in loan for use to our team, working at *SST-Laboratory* of Bari Polytechnic, formed by four members:

- Head Eng. Prof **Cataldo Guaragnella**;
- PhD Eng. **Matteo Palieri**;
- **Antonello Longo** graduated in Computer Science;
- Myself **Germano Pansini**, Electronic and Telecommunication bachelor's degree student.

My work started from the selection and then integration of all the components required for the hardware platform to begin robot's navigation tests in a complex environment.

Firstly, we are going to focus on its Chassis Structure and Locomotion Subsystem, then on the Power System, including a discussion about common Battery type suitable for mobile robots, on Motor Driver and then on Microcontroller and Sensors.

Next chapter focuses on Navigation Tests and first experimental results.

4.2 Rolling Chassis and Locomotion subsystem

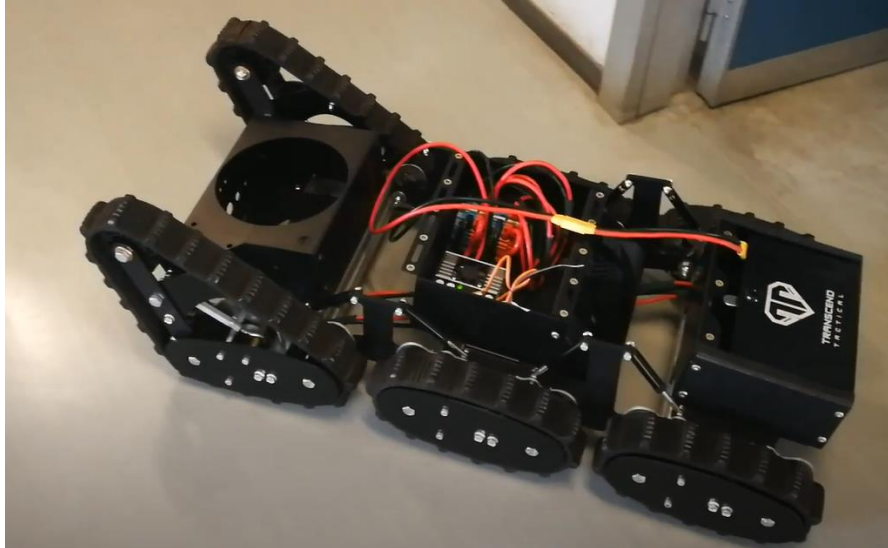


Figure 17 Picture of the robot during first Navigation test at the SST-Lab Laboratory

This type of robot is a **rolling** robot that has a skid locomotion system consisting of three different pairs of continuous tracks for each side, made with soft belts of synthetic rubber reinforced with steel plates. The body's structure of this robot is composed by three different strong metal blocks, each one with two pairs of tracks, allowing it to climb stairs and overtake objects still keeping contact with the ground with, at least, one part of its body at any time of its navigation, avoiding it to stop even in complex positions, frequent in rough and loose terrain.

This robot is aimed for First Responders, Robot exploration or Military, so its structure needs to be able to work on complex environments, climbing clothes, rocks, stairs and never being stopped by objects.

For the reason above, its chassis is really heavy and hardly damaged, allowing it to carry even more than +100lbs of payloads and guaranteeing long-life. One tracks' pair for the back of the body, second pair for the middle one and the last one for the front.

The continuous bands of tracks' plates are driven by two wheels for the back and middle block, whereas by three for the front one, in order to raise up the first part of the robot that encounters obstacles, making it easier to cling to higher objects, as stairs, as we can see in the next photo taken during one navigation test in the garden of Polytechnic of Bari.



Figure 18 Vantage climbing stairs in the garden of Bari Polytechnic

The prominent treads of the metal plates are both hard-wearing and damage resistant, furthermore, the aggressive treads of the tracks provide good traction in soft surfaces and with the rubber pads we avoid damaging paved surfaces or deforming its own wheels.

Cfr. (Wikipedia, Continuous Track, s.d.)

We're not going to discuss wheel geometry or robot's stabilisation.

The six continuous belts roll because of wheels rotation driven by six different 12V DC Motors from *AndyMark*, also connected with six individual Mini Circuit Breakers (MCB) from *Snap Action*, which are automatically operated electrical switch, intended to prevent damage to the motors as a result of excess current above the maximum theoretical threshold of 20 A, far below the stall motors current of 63.8A to be in work's safe place and avoid malfunctions. During a temporary overload or short circuit (SC), the MCB trip to protect against these possible electrical faults and equipment failure. (Motor and MCB in the photo below)



Figure 20 AndyMark 9015 (AM-0912)

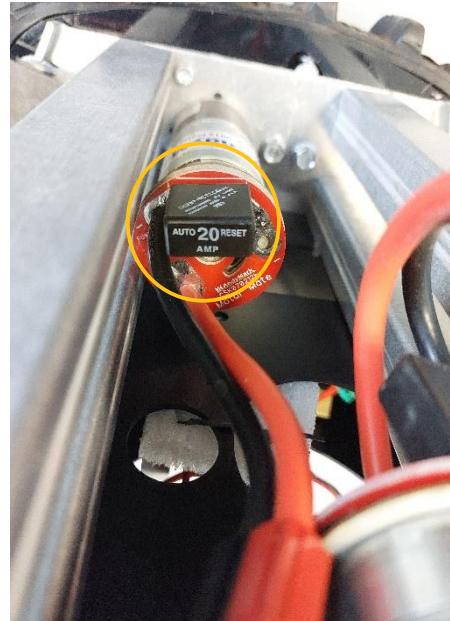


Figure 19 Snap Action Mini Circuit

As said before, wheels are driven by these DC Motors and these, in turn, are driven by the Motor Driver, powered by LiPo batteries, on which we are focusing on the following subsection, that's about the power system development.

4.3 Power System

The first issue we encountered during the internship was about selecting one battery that would have powered the Motor Driver and thus also the 6 DC Motors, thanks to a custom terminal block distributing power to them, allowing wheels rotation and so robots first movements, and another one to power the computer and sensors onboard, such cameras and lidar we'll focus on next. So, there are two different power system, that works independently, one located in the front block and the last between a wire connection of the middle and the back blocks of the robot's body, we show in the pictures:

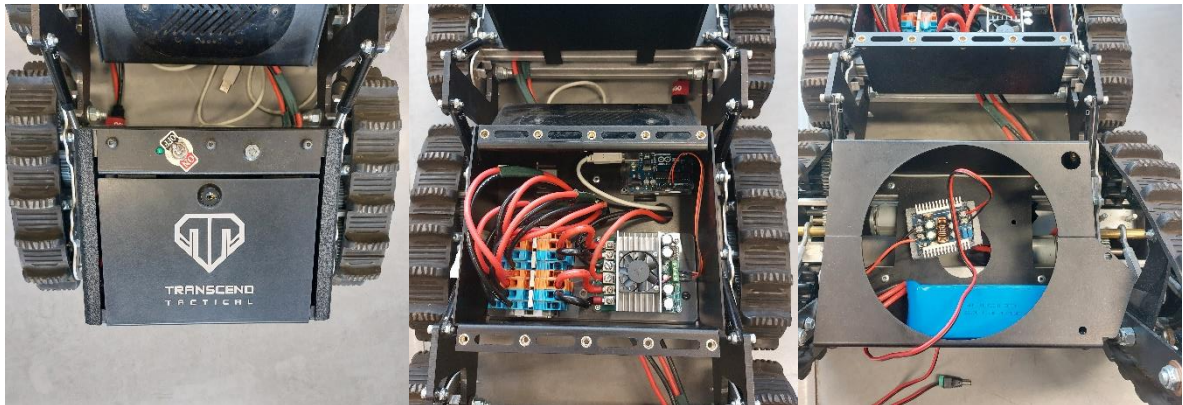


Figure 23 Back Block of the Robot

Figure 21 Middle Block of the Robot

Figure 22 Front-Block of the Robot

The Motor Driver used is *Sabertooth dual 60A* Motor Driver, which is one of the most versatile, efficient and common to find on the market. In order to power motors, in fact, we can directly connect to a battery but if we want to control these motors using other controller devices such as microcontrollers, wireless systems or more complex ones, we need Motor Drivers acting as an interface between motors and the control circuits.

When selecting the battery type, we have to pay attention to the recommended power sources section in the Motor Driver's documentation.

From the documentation of the Sabertooth 2x60 we have:

Recommended power sources are:

- 5 to 20 cells high capacity NiMH or NiCd
- 2s to 8s lithium ion or lithium polymer. Sabertooth motor drivers have a lithium battery mode to prevent cell damage due to over-discharge of lithium battery packs.
- 6v to 30V high capacity lead acid
- 6v to 30V power supply (when in parallel with a suitable battery).

Figure 24 Power Source documentation of the Motor Driver Sabertooth 2x60

Because of the wide range of actuators and electronics which go into a robot, choosing the right battery may not be an easy task.

There are many different types of batteries, but let's see the most common type suitable for robots, we can easily find in the market.

Battery type	NiMH	Li-Ion/Li-Poly high energy type	Li-Ion/Li-Poly high current type	Li-Ion/Li-Poly high safety type
Chemistry	NiMH	LiCoO ₂ LiNiMnCoO ₂	LiMn ₂ O ₄	LiFePO ₄
Nominal cell voltage	1.2V	3.6 - 3.7V	3.7 - 3.8V	3.2 - 3.3V
Operating cell voltage range	1.0 - 1.4V	3.0 - 4.2V	3.0 - 4.2V	2.5 - 3.65V
Max voltage for charging	1.4 - 1.6 V	4.2 - 4.3V	4.1 - 4.2V	3.65V
Max charging current*	0.1C or 1C (only with $\Delta V / \Delta T$)	~1C	up to 3C	up to 4C
Max discharging current	1C - 30C	1C - 2C	5C - 30C	1C - 40C
Charging method	CC with timer or $\Delta V / \Delta T$ (faster)	2 stages: CC and then CV	2 stages: CC and then CV	2 stages: CC and then CV
Specific energy	40 - 120Wh/kg	150 - 250Wh/kg	100 - 150Wh/kg	90 - 120Wh/kg
Specific power	100 - 1000 W/kg	100 - 400W/kg	400 - 5000 W/kg	200 - 7000 W/kg
Internal series resistance for a single cell	5-50 m Ω (for 18650 size)	15-100 m Ω (for 18650 size)	10-50 m Ω (for 18650 size)	6-60 m Ω (for 18650 size)

Figure 25 Battery type Table for Mobile Robotics

Despite the large variety, we have shown only the most used in robotics, with particular focus on the selected one for this project: the Lithium-Ion Polymer battery, which people also call Li-Po battery in jargon.

“Li-Po is a rechargeable battery of lithium-ion technology using a polymer electrolyte instead of a liquid one.

High conductivity semi-solid polymers form this electrolyte. These lipo batteries provide a higher specific energy than other lithium-battery types. It is a newer type of battery now used in many consumer electronics devices. They have been gaining in popularity in the radio control industry over the last few years and are now the most popular choice for anyone looking for long run times and high power.”
(Wikipedia)

The chosen batteries are being represented in the next pictures.

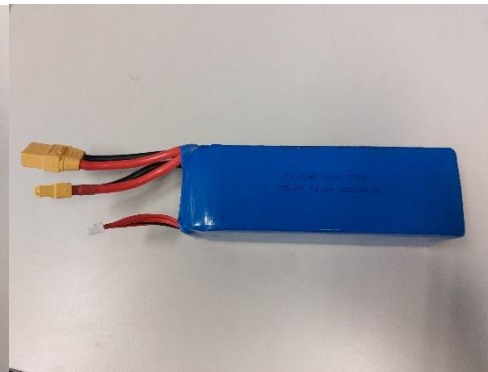


Figure 26 Shida LiPo battery for Locomotion Figure 27 LiPo battery for Computer onboard

Both batteries have same JST-XH Balance connectors but different power connectors: the first has the anti-spark XT-90S and the second one has both XT-90 and XT-60, allowing two different connection's combinations.

The power connector is the main plug we use to charge/discharge the battery, while the balance connectors is connected to each cell to

balance the charge each cell up individually, bringing their voltage up to an even level.

We are going to discuss the cell configuration soon.

As we can see from the photos, on the battery top, generally the following parameters are represented:

- Cell configuration
- Battery Voltage
- Battery Capacity
- Discharge-Rate
“C Rating”

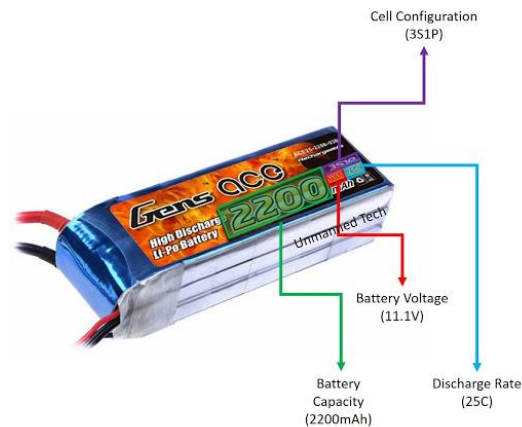


Figure 28 LiPo Battery Parameters

4.3.1 Cell Configuration

Cell configuration stands for how the cells are connected inside your battery pack. As a matter of fact, this type of battery is constructed from rectangular cells, a battery themselves, connected together to form the main battery. The single cell holds a nominal voltage of 3.7V and by connecting together in **series** we can increase the total battery voltage: for a 2-cell battery the voltage can increase to 7.4V, 14.8 for a 4-cell battery and so on. If we connect them in **parallel** instead, we increase the total capacity, the total amount of charge delivered by a fully charged battery under specified conditions of temperature and load.

Often, we can find the connection type, written as a code like *4S2P*, which represents the cell configuration itself, meaning that we have 4

cells set in series, and 2 connected in parallel, giving a total number of 6 individual cells.

So, the number of cells inside a Li-Po is what defines the voltage and the capacity of the battery.

Higher voltage means that we can drive bigger motors, whereas the duration of the battery providing energy is defined by the battery capacity.

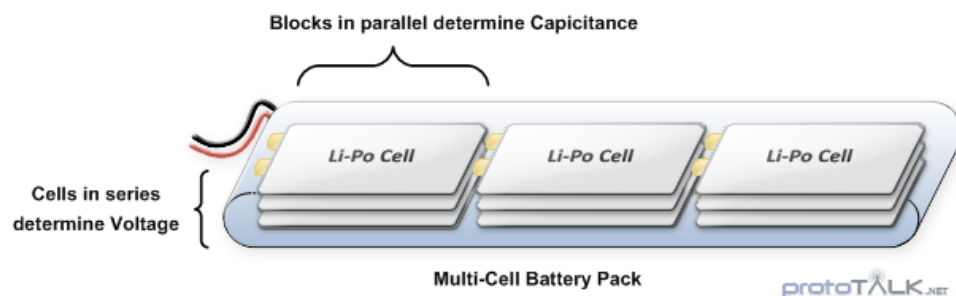


Figure 29 Cell Configuration

It's also important to have a battery tester to check, connecting the balance connector, important parameters such as battery capacity, so we know when recharge is required, or the voltage gap between highest cell voltage and lowest, allowing us to understand if we need a battery replacement.



Figure 30 Smart Battery Meter from Jrelecs checking cells voltage

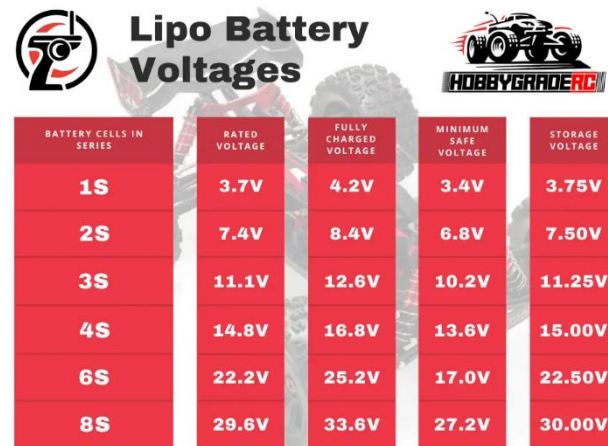
4.3.2 Battery Voltage

As we've seen above, a single cell has a nominal voltage of 3.7V. To drive 6DC Motors we've chosen a 22.2V battery. We can simply calculate how many cells are packed in series into our battery, just doing:

$$\text{Cells} = \frac{22.2V}{3.7} = 6s$$

The Voltage of a Li-Po battery pack is the amount of electrical potential a battery holds, measured in volts. (What are volts and why do they matter?, s.d.) The voltage of a battery directly influences the motor's torque, and the torque sets the RPM value (Revolutions Per Minute), a measure of speed our electric motor is operating at.

So, we need to know the electric motor's model when we have to select a lipo battery, because voltage has an impact on motors, and motors directly influence the speed.



The chart is titled "Lipo Battery Voltages" and features a red background with white text. It includes a logo on the left and a small image of a car on the right. The chart is organized into five columns: "BATTERY CELLS IN SERIES", "RATED VOLTAGE", "FULLY CHARGED VOLTAGE", "MINIMUM SAFE VOLTAGE", and "STORAGE VOLTAGE". The rows represent different series counts from 1S to 8S.

BATTERY CELLS IN SERIES	RATED VOLTAGE	FULLY CHARGED VOLTAGE	MINIMUM SAFE VOLTAGE	STORAGE VOLTAGE
1S	3.7V	4.2V	3.4V	3.75V
2S	7.4V	8.4V	6.8V	7.50V
3S	11.1V	12.6V	10.2V	11.25V
4S	14.8V	16.8V	13.6V	15.00V
6S	22.2V	25.2V	17.0V	22.50V
8S	29.6V	33.6V	27.2V	30.00V

Figure 31 LiPo Voltage Chart from hobbygraderc.com

The higher voltage is, the higher power P of the motor is, and here is the formula: $P = V \cdot I$, where "P" is power, "V" is voltage, "I" is current.

For Motor Driver's Battery: $P = 22.2V \cdot 10A = 222W$

For the Onboard-Computer Battery: $P = 22.2V \cdot 5A = 111W$

4.3.3 Battery Capacity

The value "10.000mAh" or "5.000mAh", written on our batteries, stands for the battery capacity.

Battery capacity is defined as "the total amount of electricity generated due to electrochemical reactions in the battery and is expressed in ampere hours", so "is the amount of energy that can be withdrawn from it at a particular constant current, starting from a fully charged state." ("A Lipo Battery Guide to Understand Lipo Battery"), ("Nominal Battery Capacity").

The battery capacity represents the maximum amount of energy that can be extracted from the battery under certain specified conditions. However, the actual energy storage capabilities of the battery can vary significantly from the "nominal" rated capacity. For our batteries, we have:

$$10000mAh = 10A (10Ah)$$

$$5.000mAh = 5A (5Ah)$$

For the first one we can deliver 10A to a 22.2V device for one hour, the second 5A to a 22.2V for the same amount of time.

Generally, capacity can determine how long we can run before we have to recharge it.

4.3.4 Discharge Rate (C RATING)

The C rating stands for how fast the battery can be discharged safely, it's a function of the time in which it takes to fully discharge the battery. It's defined as *the steady current in amperes (A) that can be taken from a battery of defined capacity (Ah) over a defined period of time (h)*. (Low Discharge Rate, s.d.)

For a battery with a discharging rate of 20C and a capacity of 10Ah, this equates to a discharge current of 200A in 1 hour.

4.4 Motor Driver

In this chapter we are explain what a motor driver is and how we chose one for our purpose. A Motor Driver is a breakout board that act as an amplifier-interface between motors and the control circuits, turning low current signal to high signal for driving the motors.

In our project we want to control speed and direction of the robot using Arduino as a Microcontroller, but we cannot directly connect our motors to Arduino as they do not get sufficient power from its pin. So, we basically connect the motors to the Motor Driver and then supply power to the motors using a battery through the Motor Driver.

In the next picture we can see the integration of the Motor Driver, whose output are connected to the terminal block that is distributing power to the six different motors and whose input are controlled by Arduino using **simple serial** byte commands, we're going to explain later.

We chose the Sabertooth 2x60A, which can control two different motors with output current of 60A per channel with possible peak of 120A for few seconds. (Remind we've integrated also Mini Circuit Breakers to avoid peak current going directly to the motors).

We connected the electrical terminal block that split powers through motors to the M1A, M1B, M2A and M2B, who are respectively the positive and negative terminals of Motor 1 and Motor 2. The Sabertooth in fact, is called a "Dual Motor Driver"; for this reason, we used the electrical terminal block to drive more motors at the same time with the same Driver.

Then we connected our 22.2V LiPo battery (10Ah) to the terminals B+ and B-, interposing a toggle switch in the positive wire to avoid possible sparks between connectors plug, being careful to respect the right polarity in wiring cables. The toggle switch is accessible from the back block of the robot, allowing us to power off the robot in case of its wrong behaviours.



Figure 32 Motor Driver controlled by Arduino to drive Motors

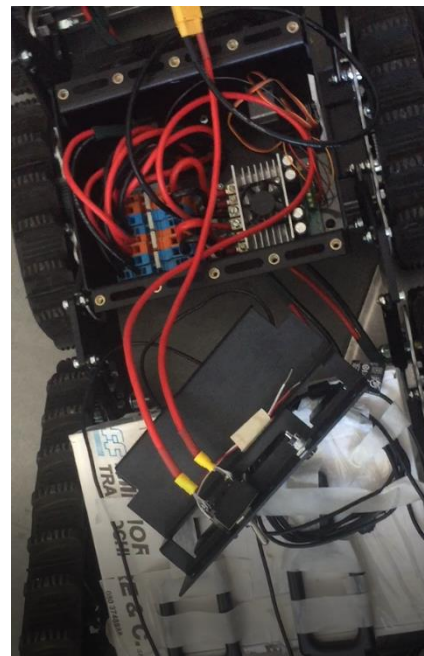


Figure 33 Toggle Switch implementation

It's also possible to power Arduino directly from the 1A switching BEC (Battery Elimination Circuit) of the Motor Driver, which converts main LiPo battery pack voltage to a lower one, 5V in this case, for electronic devices. The input signal coming from Arduino's pin, that actually control the Sabertooth, is connected to terminal S1.

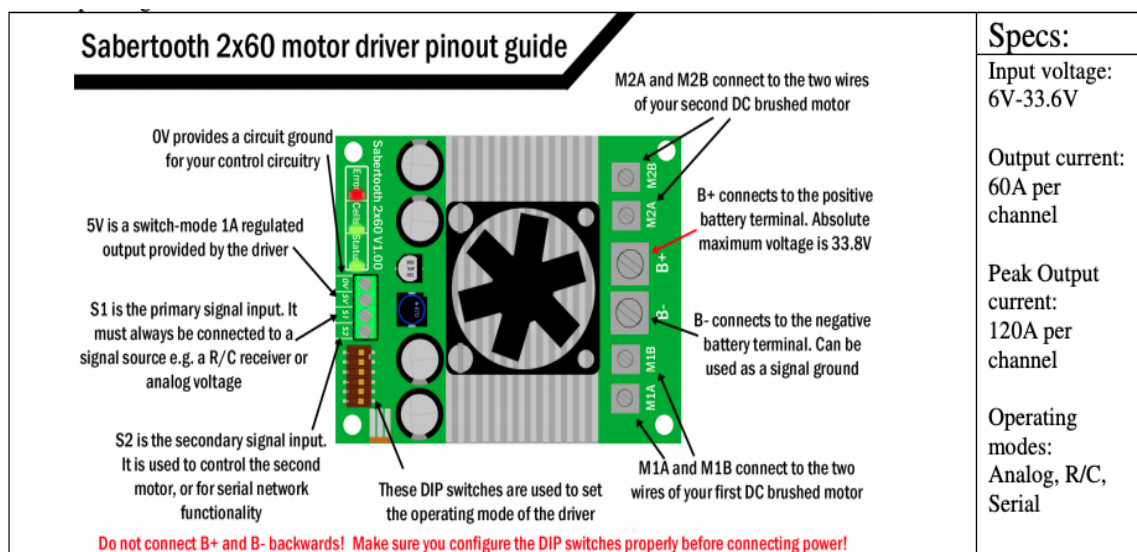


Figure 34 Sabertooth 2x60 Motor Driver Pinout Guide from Dimension Engineering

This Motor Driver has overcurrent and thermal protection and has a regenerative topology that means our batteries get recharged whenever we command our robot to slow down or reverse. Furthermore, has a lithium cut-off technology that allows Sabertooth to operate safely with lithium battery: it automatically detects the number of lithium cells at start-up and sets a voltage cut-off of 3.0V per cell, so they operate in safe mode preventing possible damage to the battery.

This device can also have a wide variety of operating modes, just changing its DIP switches, but we're explaining only ours.

We set the DIP Switches in this way, shown in the next picture:

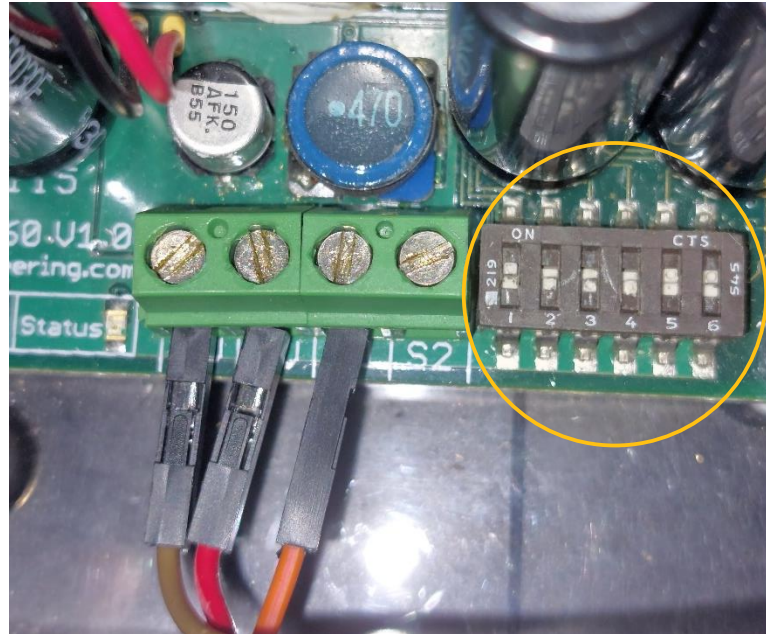


Figure 35 DIP switches setting Operating Mode of the Motor Driver

- 1) Switch **1 ON** and Switch **2 OFF** are setting the Simplified Serial communication between Arduino and Motor Driver
- 2) Switch **3 OFF**: set Lithium cut-off mode to automatically detect the number of series lithium cells and set a cut-off voltage of 3.0V per cells
- 3) Switch **4 OFF**: the Motor Driver is in Independent Mode
- 4) Switch **5 ON**: the response to input signal will be linear
- 5) Switch **6 ON**: set the input signal range to 0V to 5V with a zero-point of 2.5V

4.5 Microcontroller (ARDUINO UNO R2)

To be able to have an interface between the Motor Driver and motors we chose the Arduino Uno R2, the brain of our robot locomotion subsystem.

The Arduino is powered from the Sabertooth via BEC (Battery Elimination Circuit), providing the common voltage value for electronics devices of 5V DC.

We connected 0V from the Sabertooth to Arduino GND, that stands for ground, S1 to Arduino TX, and 5V to Arduino Vin, the input voltage to the microcontroller's board when using an external power source.

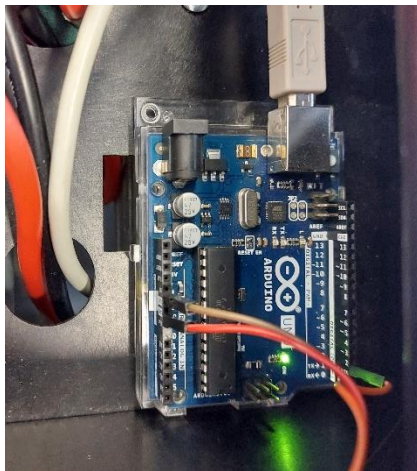


Figure 36 Arduino UNO R2

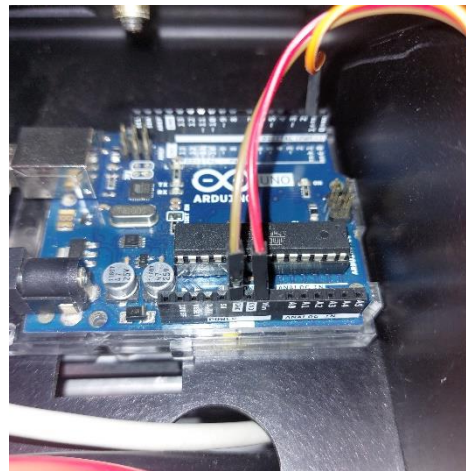


Figure 37 Arduino Pin connection

The USB port is connected to the fanless computer onboard, to receive or transmit data, such as the joystick position in the first guided navigation tests.

Once we have implemented the Sabertooth library called “SabertoothSimplified” we can allow the communication system

between Arduino and the Motor Driver with Simplified Serial protocol.

4.5.1 CODE for Simplified Serial Communication

Serial communications provide an easy and flexibly way for our Arduino board to interact with our computer and other devices.

Serial communication is a method of sending and receiving information one bit at a time. This method takes many different forms: there's many ways to serially communicate from one device to another. The most familiar method of serial communication is over USB (Universal Serial Bus), one of serial communication protocols.

Arduino it's equipped to communicate using different protocols like:

- Serial Peripheral Interface (SPI);
- Inter-Integrated Circuit (I2C);
- TTL (Transistor-Transistor logic) Serial using U(S)ART (Universal Synchronous/Asynchronous Receiver/Transmitter)

The last mentioned is the protocol is what we are using to set speed and direction of the motors and the code was developed with help of my teammate Antonello Longo, so we are not going to explain it, but is shown in the next picture.

We only have to know that Arduino is receiving joystick's position data from the fanless computer onboard, that is receiving from another computer, the main remote station of our laboratory, and is converting this type of code in voltages value to the motors to set speed and direction. Serial hardware sends and receive data as electrical pulses. Zeros and ones that carry the information that makes

up a byte can be represented in various way. The scheme used by Arduino is 0V to represent a bit of value of 0, and 5V (or 3.3V) to represent a bit value of 1. This is referred to as the TTL level (Transistor-Transistor Logic)

```

1 #include <SabertoothSimplified.h>
2
3 #define motor_full_speed_fw 127
4 #define motor_full_speed_bw -127
5
6 #define motor_smoothed_speed_fw -25
7 #define motor_smoothed_speed_bw 25
8
9 /*
10 #define slow_forward_right 65
11 #define slow_forward_left 193
12 #define slow_backward_right 63
13 #define slow_backward_left 191
14
15 #define fast_forward_right 127
16 #define fast_forward_left 255
17 #define fast_backward_right 1
18 #define fast_backward_left 128
19 */
20
21 #define stop_motors 0
22
23 String readS;
24 SabertoothSimplified ST;
25
26
27 void setup() {
28     // put your setup code here, to run once:
29     Serial.begin(9600);
30     pinMode(13,OUTPUT);
31     SabertoothTXPinSerial.begin(9600);
32 }
33
34 void loop()
35 {
36     while (Serial.available()){
37         delay(1); //delay to allow buffer to fill
38         if (Serial.available() >0){
39             char c = Serial.read(); //gets one byte from serial bu
40             readS += c;
41         }
42     }
43
44     if (readS.length() >0){
45         if(readS == "0"){
46             stop();
47         }
48         else{
49             if(readS == "1"){
50                 digitalWrite(13,HIGH);
51                 delay(50);
52                 digitalWrite(13,LOW);
53                 move_forward();
54             }
55             else{
56                 if(readS == "2"){
57                     digitalWrite(13,HIGH);
58                     delay(50);
59                     digitalWrite(13,LOW);
60                     move_right();
61                 }
62             }
63         }
64         else{
65             if(readS == "3"){
66                 digitalWrite(13,HIGH);
67                 delay(50);
68                 digitalWrite(13,LOW);
69                 move_backward();
70             }
71             else{
72                 if(readS == "4"){
73                     digitalWrite(13,HIGH);
74                     delay(50);
75                     digitalWrite(13,LOW);
76                     move_left();
77                 }
78             }
79         }
80     }
81 }
82
83 readS = "";
84 Serial.flush();
85 /*
86 if(readString == "1"){
87     move_forward();
88 }
89 elif()
90 Serial.print("Arduino received: ");
91 Serial.print(readString); //see what was received*/
92 }
93
94 void move_forward(){
95     ST.motor(1,motor_smoothed_speed_fw);
96     ST.motor(2,motor_smoothed_speed_fw);
97 }
98
99 void move_right(){
100     ST.motor(1,motor_smoothed_speed_fw);
101     ST.motor(2,motor_smoothed_speed_bw);
102 }
103
104 void move_left(){
105     ST.motor(1,motor_smoothed_speed_bw);
106     ST.motor(2,motor_smoothed_speed_fw);
107 }
108
109 void move_backward(){
110     ST.motor(1,motor_smoothed_speed_bw);
111     ST.motor(2,motor_smoothed_speed_bw);
112 }
113
114 void stop(){
115     ST.motor(1,0);
116     ST.motor(2,0);
117 }

```

Figure 38 Code for Simplified Serial communication between Arduino UNO and Sabertooth 2x60

4.6 Power system for the onboard Computer

We've dedicated the front metal block of our robot to the power system allocation of its brain, a fanless computer.

The fanless computer is powered onboard from our 22.2V 5000mAh LiPo battery and gives the machine its intelligence layer and its ability to perform tasks.



Figure 39 Fanless computer used

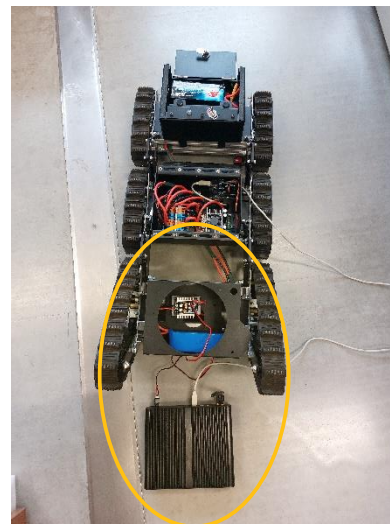


Figure 40 Computer Power system

After finding out the required input voltage of our fanless, 12V DC, we implemented a DC-DC Converter to manually, thanks to its trimmer, set the output voltage of our LiPo battery to that value.

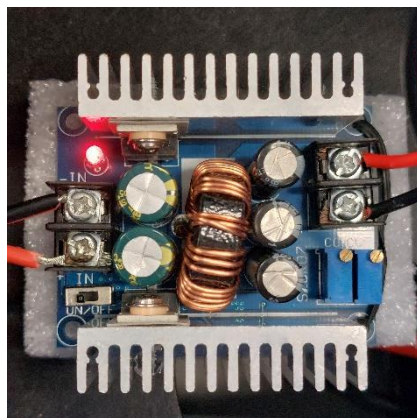


Figure 42 DC-DC Converter

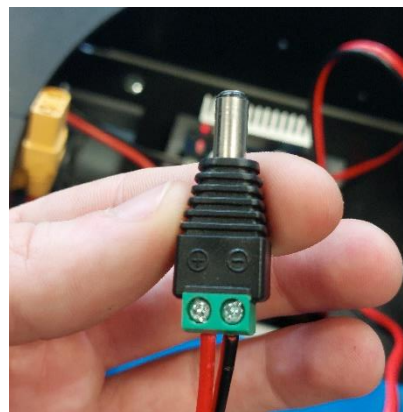


Figure 41 Male Plug Power Jack



As we step-down AC voltage using transformers, we can also step-down DC voltage using DC-DC Buck converters.

A Buck converter outputs a lower voltage than the original voltage, while a Boost Converters supplies a higher voltage. For our purpose we have to step-down the 22.2V supplied by the LiPo battery to the input voltage of 12V required to power the computer, using a male power jack adapter plug (2.1mm x 5.5 mm). This is done by chopping the higher input voltage up by rapidly switching the output power transistor on and off so that the output essentially looks like a square wave and then using a filter to smooth it back into a DC voltage on the output of the filter.

The efficiency of DC-DC Converters is typically much better than the equivalent linear regulator circuit.

To be sure to output the required voltage to power our fanless computer, I've taken voltage measurement at the plug power jack with a classical Multimeter.

4.7 Integration of LiDAR L515

The main vision sensor we've used for the Navigation test of our robot is the LiDAR L515 from *Intel RealSense*. This is an ultra-compact high-resolution depth camera with Intel's original MEMS (Micro Electromechanical System) mirror scanning technology, which represents one of the solid-state systems. This LiDAR is designed for indoor applications that require depth data at high resolution and accuracy. The minute three-dimensional structure of MEMS can handle a wide variety of input and output signals while consuming low power.

It has been incorporated in various research and development such as telepresence robots and automatic guided vehicles (AVGs) that are specialized in the automation of transportation and transportation operations, with its only 100g of weight.

(Solid-State LiDAR self-positioning and map generation for mobile robots (RealSense), 2021)



Figure 43 L515 from IntelRealSense

LiDAR is the acronym for Light (Laser) Detection and Ranging and is a device that uses a particular method for determining range, variable distance, that consists of targeting an object or a surface with a laser and measuring the time for the reflected light to return to the receiver (TOF: Time-of-flight).

Light is emitted at time t_0 , hits an object, is reflected back and then measured by an array of sensors at time t_1 . The measured interval can be easily converted into a precise distance based on knowledge about the speed of light.

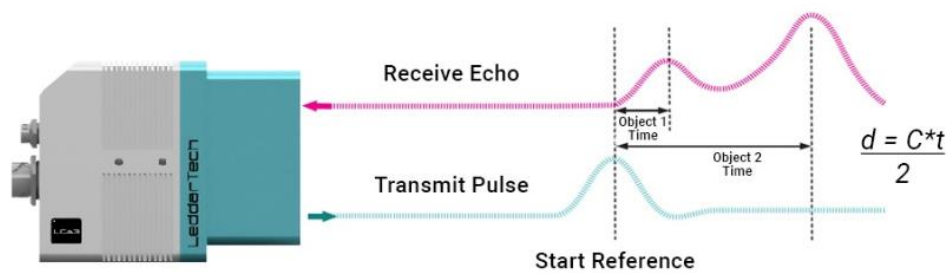


Figure 44 Time of Flight method for LiDARs

Efficient LiDARs can process more than 1 billion samples per second, so it is easy to see how these systems can also determine an object's shape, size and movements in the surrounding environment.

Unlike "normal" cameras, they function independently of the ambient lighting and can achieve fantastic results day and night without any loss of performance due to disturbances such as shadows, sunlight or headlight glare.

They can recognize different objects because of their different emission characteristics, as the quantity of light reflected or absorbed. In fact, it's a common sensor that is used in control and navigation for autonomous robots, thanks to the vast number of measurement points in LiDAR technology that allows a very accurate 3D reconstructions

of a scene. So, every light pulse emitted provides specific information about the relative distance and size of the detected object and allows the system to create a precise three-dimensional mapping of the environment. This can also be achieved with cameras, but image analysis is much simpler and faster with LiDAR and furthermore postprocessing techniques with cameras have to be applied to the 2D images to determine size and distance, not to the measurements themselves, determining less accuracy.

We power the L515 LiDAR directly from our fanless computer onboard via USB 3.0, that support the common voltage for electronics of 5V DC, integrating it in our hardware platform for robot navigation using ROS Melodic middleware running on Ubuntu.

The data transmitter for the LiDAR is the USB cable itself.

4.8 ROS and SSL-SLAM

We used our sensor in combination with the **Robot Operating System, ROS**, that runs on the fanless computer onboard.

ROS is a set of libraries and tools that help us building robot applications such as **SSL-SLAM** (Solid-State LiDAR Simultaneous Localization and Mapping), on which we have worked. SSL-SLAM is an open-source code, available to anyone on GitHub.

Light detection and ranging plays a key role in SLAM and are considered nowadays one of the most important perceptual devices, which is also becoming cost-effective and lightweight solution.



Solid-state LiDAR has higher scan frequency and angular resolution compared to mechanical LiDAR, but the limited Field-of-view (FOV) creates uncertainty with the existing LiDAR SLAM algorithm we this new sensing device requires a more robust and computationally efficient SLAM method. For this SSL-SLAM has been proposed as a new SLAM framework, that's include feature point extraction, map construction and odometry estimation.

Thanks to the L515 working on ROS, that runs directly onboard, while the robot navigates, we can construct and continuously update a point cloud map of an unknown environment while simultaneously keeping track of the agent's location within it.

Although we have worked on this software side too, I'm not going to explain it deeply such it is a wide and complex layer that would have required a thesis itself, but I'm only showing in the next dedicated section the experimental results of the reconstruction of unknown and complex environment.

4.9 Joystick and guided Navigation

To perform first navigation tests, we controlled robot from a *Logitech* joystick connected to a remote station of our laboratory.

Thanks to the program we've seen in *Figure 24*, that my teammate Antonello Longo and I developed, Arduino receive data of the joystick position and convert it to voltage value, manageable for the Motor Drivers, talking to the motors.

But before sending to Arduino, the joystick position data are converted by a Program running on ROS, that Antonello Longo developed so we're not going to deal with.



Figure 45 First guided Navigation test with the remote Joystick

From our client Station we can also have access to the fanless computer onboard via Secure Shell Protocol (SSH), that is a client-server architecture that uses public-key cryptography to authenticate the remote computer and allow it to authenticate the user.

5.0 Tests and Experimental Results

5.1 Introduction

In this chapter we now analyse the testing process and experimental results. All the tests were carried out at SST-Laboratory in both indoor and outdoor environments.

The goal of the testing phase is to evaluate and improve robot performance in:

- **On-board Powering Systems**
- **SSH Protocol**
- **SSL-SLAM**

5.2 TEST 1 - On-board Powering Systems

Once we've completed the hardware and software integration, we turned on our mechanical and electrical devices as follows:

- 1) Power Motors through Motor Driver output, enabling current flow, by placing the toggle switch on ON-mode and verify that Motor Driver's led blinks green.
- 2) Enabling current flow to the DC-DC Converter connected to our on-board Fanless Computer to power on robot brain and sensors (LiDAR L515 and possible Cameras)

5.3 TEST 2 – SSH Protocol

To enable data communication through the on-board computer and the client remote station via same Network and to be able to access remotely to the robot's computer, we selected the SSH protocol.

The SSH server wasn't installed by default on Ubuntu desktop system in the fanless computer onboard, but we did it easily from the standard Ubuntu repositories.

To install and enable SSH on our Ubuntu system we open the terminal and enter the following commands:

```
sudo apt update  
sudo apt install openssh-server
```

After entering the password and once the installation is completed, the SSH service starts automatically. To verify that installation is successful and SSH service is running, we can print the SSH server status by typing:

```
sudo systemctl status ssh
```

If firewall is enabled, we have to open the SSH port:

```
sudo ufw allow ssh
```

After we've installed SSH on the Robot Intelligence layer, we can connect to it via SSH from any remote machine.

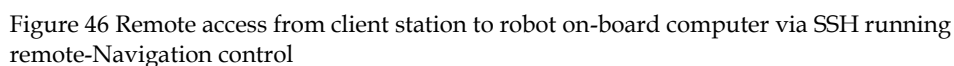
From the client remote desktop, we type:

```
ssh username@public_ip_address
```



Once we enter the password of the computer onboard, we are logged in to our Ubuntu machine. If both computers are running over the same Internet Network, SSH communication data will be transferred securely over the same Network, just by knowing the public IP address to configure our router to accept data on a specific port and send it to the Ubuntu machine where SSH is running.

The picture below shows the remote access to the fanless computer via SSH to run the Remote-Control code allowing us to guide robot's Navigation from our client station via USB Joystick.



5.4 TEST 3 - SSL-SLAM

SSL-SLAM is an open-source code that, with the use of LiDAR sensors, allows Simultaneous Localization of the robot and Mapping of the surrounding environment.

In this way, robot can at any time know its position and detect object surrounding it, while building and updating a 3D map of the unknown environment.

Before integrating the L515 from IntelRealSense in our project, we tested the SSL-SLAM code carrying the LiDAR around my own house to test it.

These are some first good results of the environment reconstruction:

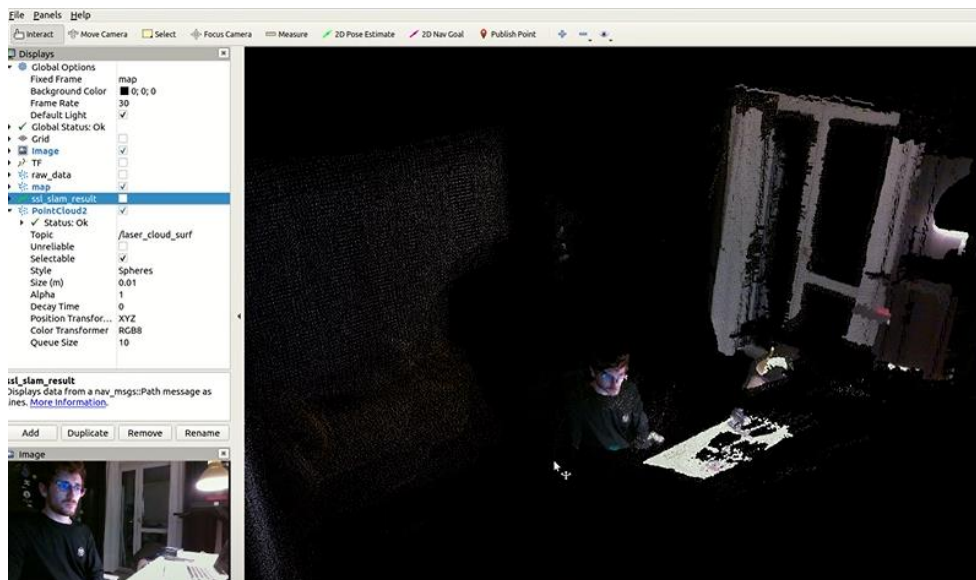


Figure 47 SLAM test with L515 in my room

Picture are taken from RViz, that is a 3-dimensional visualization software tool for ROS that helps to visualize what the robot is seeing and doing. It enables to see the robot perception of the world.

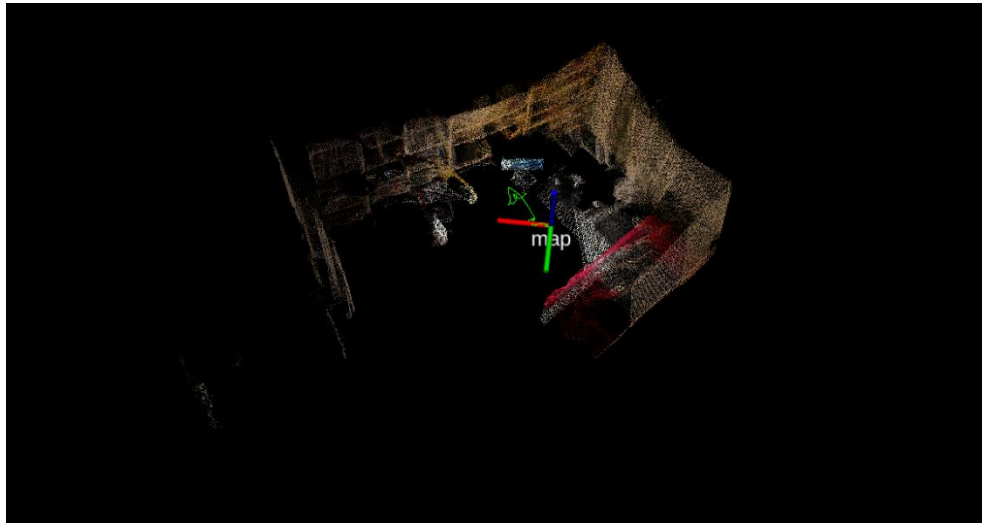


Figure 48 3D Map of my room with L515

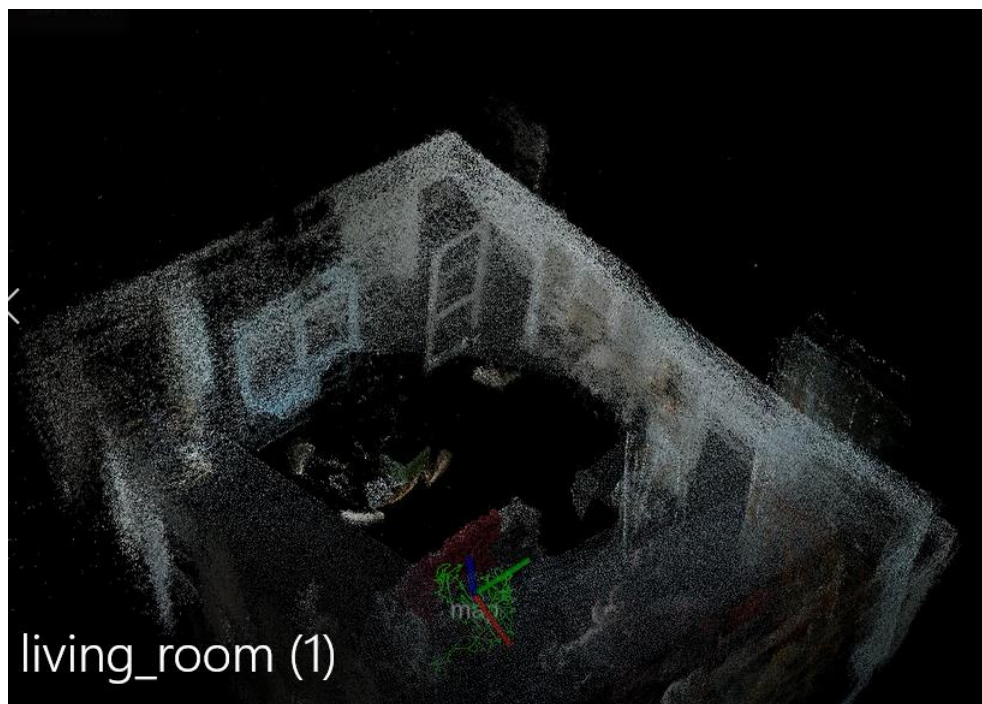


Figure 49 3D Map of my Living room with L515

After testing SSL-SLAM in my house we integrated the L515 in our robot, connecting it to the fanless computer on-board, to test 3D map reconstruction of the SST-Laboratory hallway and office, while controlling its Navigation via Joystick from the remote client station.

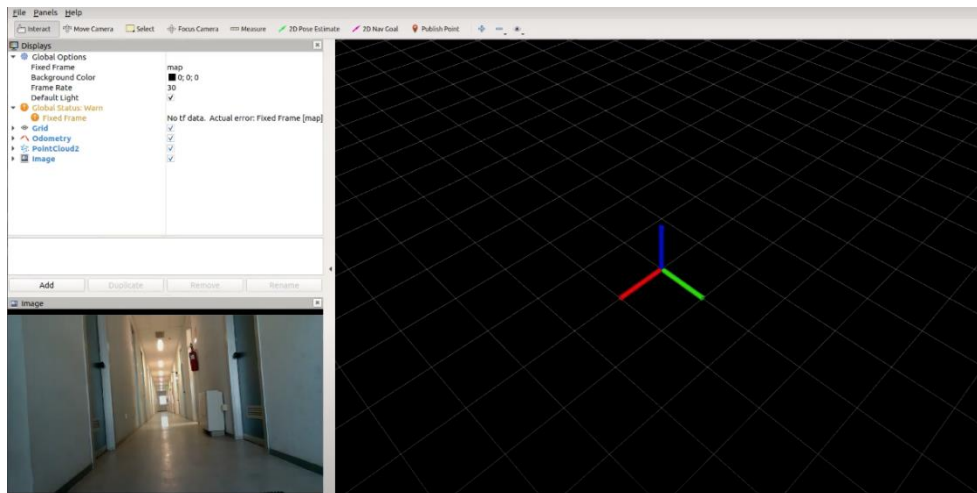


Figure 50 Robot starting SSL-SLAM in our Laboratory hallway (Step 0)

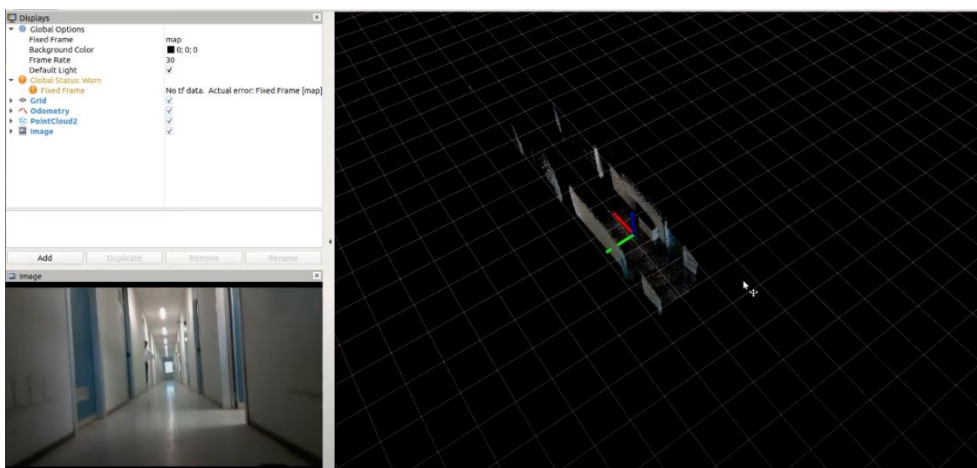


Figure 51 Point Cloud Map of our Laboratory hallway (Step 1)

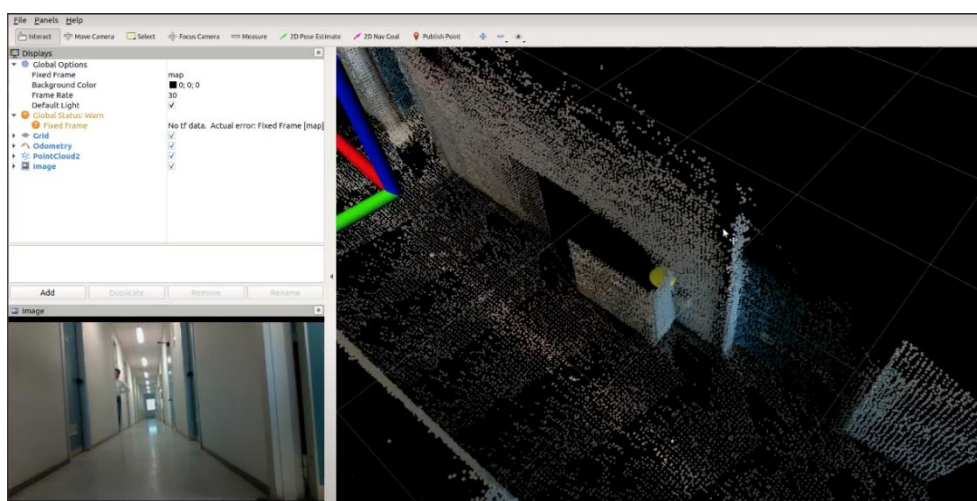


Figure 52 Zoomed Point Cloud Map of our Laboratory hallway (Step 2)

While robot is building and updating 3D map of its environment, we can also move within it and zoom the image.

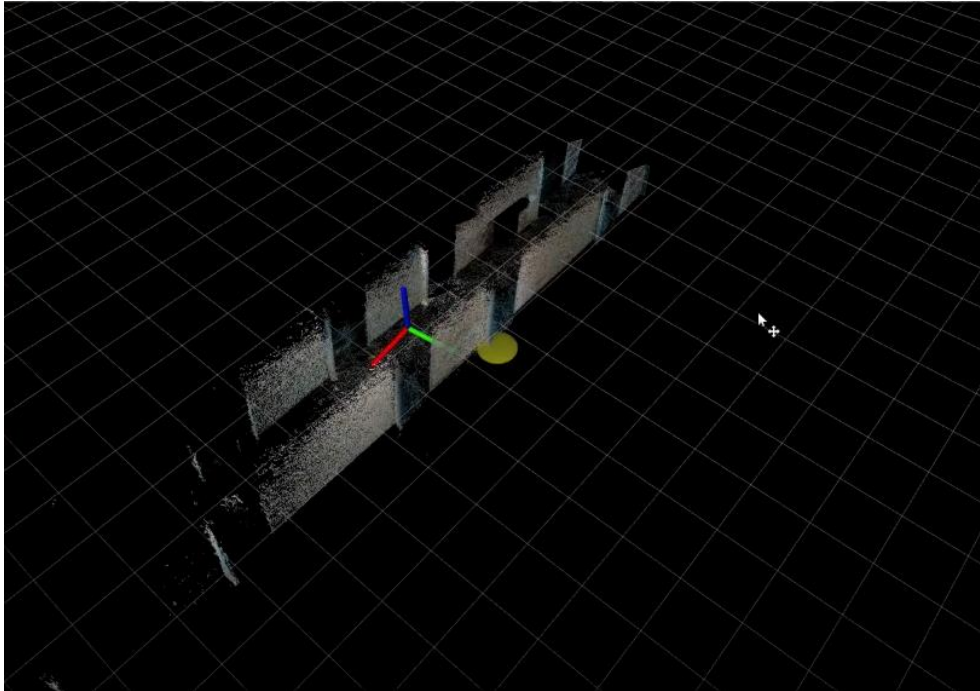


Figure 54 Final 3D map of our Laboratory hallway (Step 3)

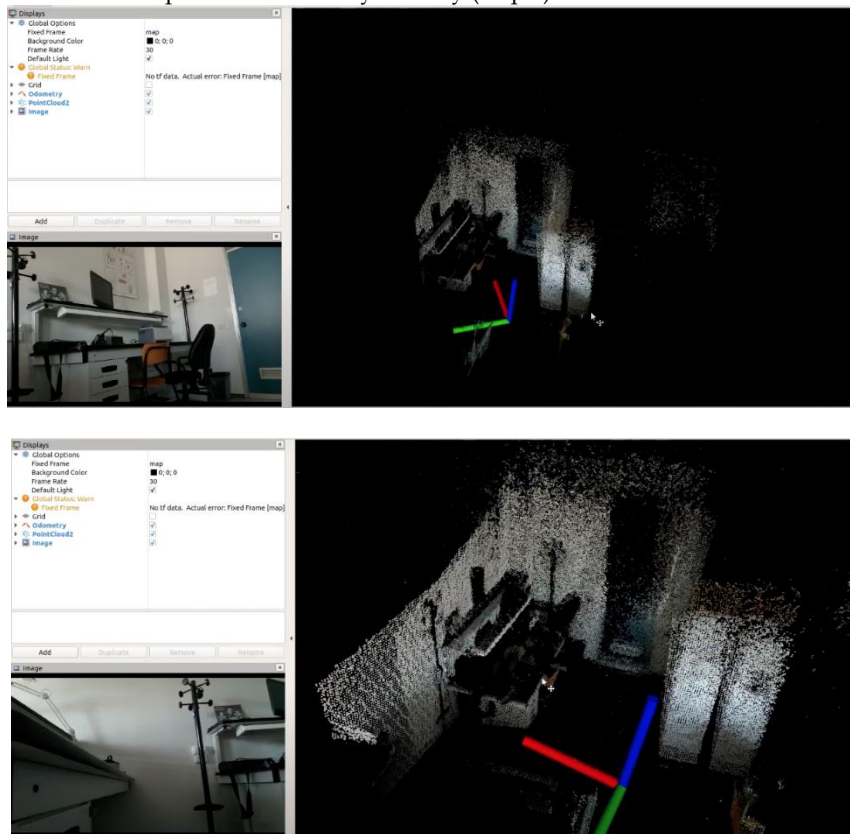


Figure 53 3D Map of SST-Laboratory

6 Conclusions and future Works

To conclude this thesis, we can say that facing real problems in designing and developing my first hardware platform for mobile robotics has allowed me to have a better understanding of what robots are and how do they work, even if all my work was based on self-study and research.

There are still many challenges we have to face, and for this reason my team is going to continue developing the custom Vantage Patrol Robot in collaboration with “E-Viewing”.

Next capabilities to achieve are:

- Obstacle Avoidance.
- Fully Autonomous Navigation.
- SSL-SLAM with a greater number of LiDARs and additional sensors.

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